

The effect of deviations from the Kerr metric on the fluorescent Fe K α line as a test for general relativity.

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ABSTRACT

The relativistically broadened Fe K α line is a key probe of the spacetime geometry around accreting black holes. While the Kerr metric is the standard description, parametric deviations such as the Johannsen metric allow observational tests of the no-hair theorem. We use the relativistic ray-tracing code *Gradus.jl* to model photon paths from a lamp-post corona to a thin accretion disc and a distant observer in a non-Kerr spacetime. This project assesses the potential of current observations to constrain deviations from the Kerr metric and to test general relativity in the strong-field regime.

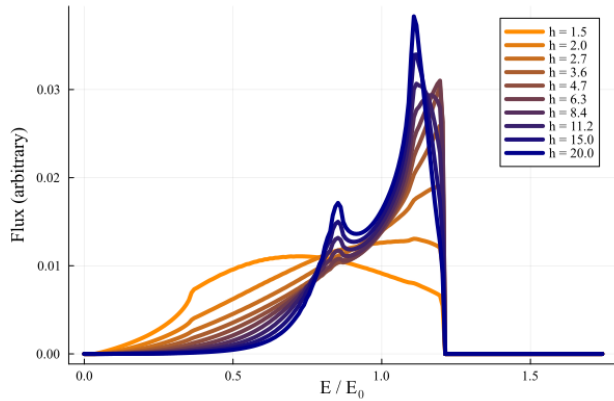


Figure 1. Line profile from Kerr black hole ($\alpha = 0.998$) created using *Gradus.jl* for a lamp post corona illuminating an accretion disc inclined by 60° to a distant observer. The corona height varies from $1.5 - 20R_g$.

1 BACKGROUND

The fluorescent Fe K α line is a prominent feature in most of the reflection spectrum of Active Galactic Nuclei (AGN) and X-ray Binaries (XRBs) and a valuable diagnostic tool in studies of these compact objects (Guainazzi et al. 2006). It arises from illumination of infalling matter in an accretion disk by X-rays from the compact corona of hot plasma near the black hole. These X-rays excite the iron in the disk leading to the distinctive emission at 6.4keV. In the rest frame of the disc this is a narrow emission line, however, the effects of general relativity cause the feature to be broadened significantly (see Fabian et al. (2000); Miller (2007); Pszota & Kovács (2023) for reviews) as exemplified in Figure.1. Material in the disc near the black hole can be travelling at significant fractions of the speed of light so the rotation of the disc at an inclination to the observer as well as Doppler beaming from the proceeding side lead to a characteristic asymmetric double peaked line (Laor 1991; Fabian et al. 1989). Near the black hole, gravitational redshift also needs to be taken into account. This makes the final observed line profile sensitive to the path light takes around the black hole. This characteristic is what makes the choice of metric important when evaluating the Fe K α line.

The standard description of a spinning object comes from the Kerr metric (Kerr 1963). It is the accepted metric used to describe particle and photon paths in near the black holes at the centre of AGN and XRBs. There does exist however other metrics which may be able to describe the inner regions of these objects. One such metric is the Johannsen metric (Johannsen & Psaltis 2011). This metric varies from the standard Kerr metric with two additional parameters ϵ_3 and α_{13} (Guo et al. 2025). When these are zero there is no difference between this metric and that of the standard Kerr but non zero values allow for small perturbations from this. It is therefore possible to use this metric as a test for the predictions of general relativity overall. The no hair theorem suggests that the only two parameters needed to define a black hole is its spin and mass but if any of these others show evidence for non zero values then it may point to this being not entirely correct.

2 PROPOSAL

In this project we will use the relativistic ray tracing code *Gradus.jl* (Baker & Young 2026) to calculate the photon paths of light emitted from a simulated corona near the black hole to the disc and then to a distant observer. *Gradus* allows for solving the geodesic equation in any given spacetime metric. We will use the Johannsen metric to solve for the light paths from a simple illuminating lamp post corona to explore the possible parameters for which observables of the black hole are most sensitive to deviations from the Kerr metric. Assuming a thin disc accretion model, this leaves a five dimensional parameter space to explore in a , h , θ , ϵ_3 , and α_{13} corresponding to the spin, corona height, inclination angle, and the two perturbation parameters from the Johannsen metric respectively. These are the parameters which can impact the observed Fe K α line profile. With the parameter space explored we would then aim to find an object from archival observations which has identified observables near this sensitivity point to explore if using a non zero ϵ_3 , and α_{13} improves the overall fit to the observed spectra. This may be done by either fitting the the Fe K α line at 6.4keV itself or by using the created line profile as a convolution kernel for a rest frame reflection spectra. A significant improvement to the fit may suggest deviations from general relativity.

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